Experiment no.

Aim: Measurement of 3 phase power by using 2 wattmeter method.

Specifications of machine: 3-Phase induction motor.

Capacity: 5 HP

Rated voltage: 400/440V

Rated current: 7A

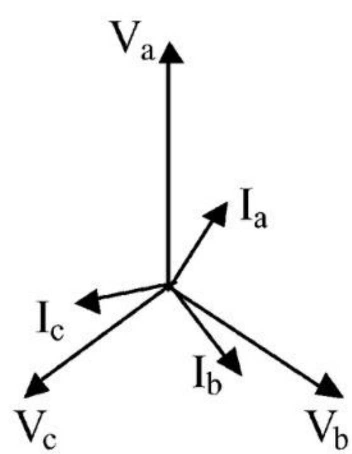
Rated Frequency: 50Hz

Rated Speed: 1500RPM

Instruments Required:

Sl.no	Name	Type	Range
1	Wattmeter (2 Nos.)	Dynamometer	5/10A, 200/400V
2	Ammeter	MI	0-10A
3	Voltmeter	MI	0-600V
4	3-Phase Variac		400/0-400V, 15A

Theory: This method of measuring power in a three-phase circuit is applicable for balanced as well as unbalanced loads. The vector diagram for balanced load is given below:



$$V_A = V_B = V_C = V$$

$$V_{AB} = V_{BC} = V_{CA} = \sqrt{3}V$$

$$I_A = I_B = I_C = I$$

$V_A, V_B\ and\ V_C$ are the vectors representing phase voltages and $I_A, I_B\ and\ I_C$ are the phase currents in phase A, B and C respectively at a lagging power factor of cosΦ.

For wattmeter W_1, the current is I_A in the current coil and the voltage across its pressure coil is V_{AB}. Therefore, its reading is

$$W_1 = \sqrt{3}VI\cos(30+\Phi) \text{ ------------------------ } (1)$$

For wattmeter W_2, the current in its current coil is I_C and the voltage across the pressure coil is V_{BC} . Therefore,

$$W_2 = \sqrt{3}VI\cos(30-\Phi) \text{ ------------------------ } (2)$$

Hence, total power read by two wattmeter is,

$$W_1 + W_2 = \sqrt{3}VI\cos(30+\Phi) + \sqrt{3}VI\cos(30-\Phi)$$

$$= 3VI\cos\Phi \text{ ------------------------------ } (3)$$

$$= 3\ x\ power\ per\ phase \text{ = Total}$$

Similarly,

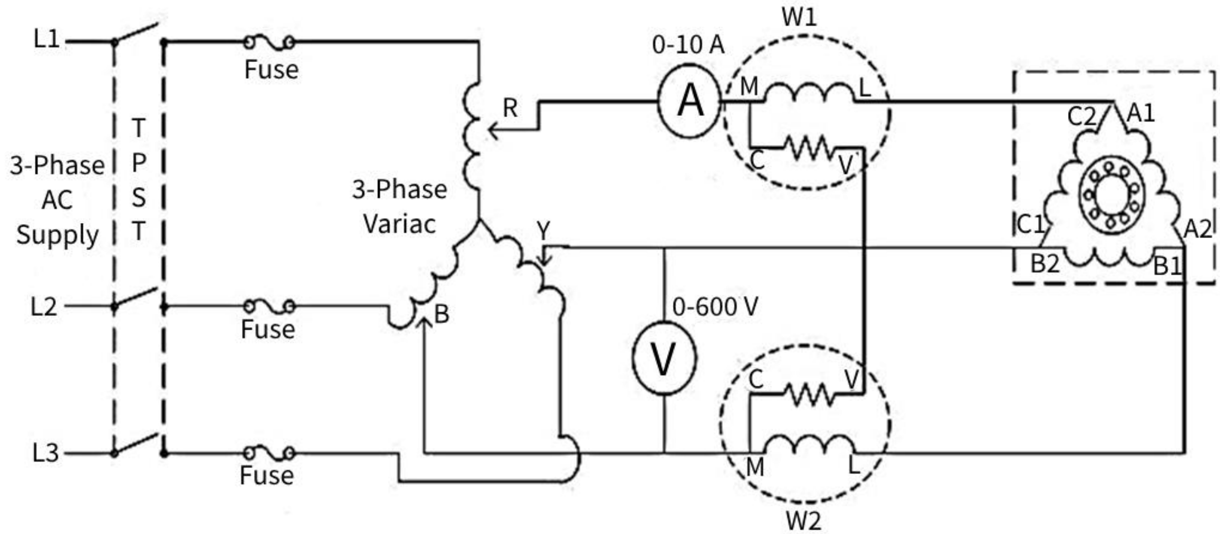
$$W_1 - W_2 = \sqrt{3}VI\cos(30+\Phi) - \sqrt{3}VI\cos(30-\Phi)$$

$$W_1 - W_2 = \sqrt{3}VI\ sin\ \Phi \text{ -------------------------- } (4)$$

From eq. 3 and 4

$$tan\Phi = \sqrt{3}\,\frac{(W_1-W_2)}{(W_1+W_2)} \text{ or}$$

$$cos\Phi = \frac{1}{\sqrt{1+tan^2\Phi}}$$



Circuit Diagram

Procedure:

1. Connections are made as per the circuit diagram.
2. Ensure that the 3- ϕ variac- is kept at minimum output voltage position.
3. Switch ON the supply. Increase the variac output voltage gradually until rated voltage is observed in voltmeter. Note that the induction motor takes large current initially, so, keep an eye on the ammeter such that the starting current should not exceed 7 Amp.
4. By the time speed gains rated value, note down the readings of voltmeter, ammeter, and wattmeter at different loads.
5. Bring back the variac to zero output voltage position and switch OFF the supply.

Observation Table:

Sl.no.	Current (in amp)	Voltage (in volt)	Wattmeter reading		W=W1+W2	tan Φ	cos Φ
			W_1	W_2			
1							
2							
3							
4							

Multiplying factor of W_1=

Multiplying factor of W_2=

Calculations: Power at different load W = ? & Power factor $cos\Phi$ =?

Result:

Precautions:

1. Connections must be made tight.
2. Before making or breaking the circuit, supply must be switched off.